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REVIEW

Examining AI Use in Educational Contexts: A Scoping Meta-Review and Bibliometric Analysis

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Abstract

With growing research attention in artificial intelligence in education (AIED), there is a profusion of systematic reviews that have investigated AI integration in a wide variety of educational contexts, including PreK-12 schools and higher education. Even though existing systematic reviews have explored effects of AI on education, few studies have synthesized the results of those reviews. To fill this gap, we conducted a scoping meta-review and bibliometric analysis to map the broad field of AIED and identify research gaps based on our analysis. Following the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines, we searched Scopus and Web of Science and collected 126 review articles from 2014 to Mid-August of 2023 that satisfied our inclusion criteria. Specifically, we used four criteria for the screening process: (1) the article's full text was available in English; (2) the article was published in peer-reviewed journals; (3) the article was a systematic review; and (4) the article was focused on AI use in one or multiple educational context(s). Our findings revealed the key characteristics of the included reviews, such as keyword co-occurrence network, co-authorship network, geographic distribution of authorship, educational levels, and subject areas. Three major themes related to AIED were generated, encompassing AI's impact on teaching, learning, and administration. We also identified commonalities and nuanced differences between PreK-12 education and higher education along with research areas that have been underexplored. Our study not only expanded understanding of studies on AIED, but also provided suggestions on how to strengthen future research.

Keywords Artificial intelligence · Education · Scoping meta-review · Teaching · Learning · Administration

Extended author information available on the last page of the article

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Introduction

With artificial intelligence (AI) technologies developing at a fanatical pace in the last decade, there has been an exponential growth in studying AI integration into education, ranging from early childhood education to higher education (e.g., Gulz et al., 2020; Ma & Siau, 2018; Moreno-Esteva et al., 2018; Ocaña-Fernández et al., 2019; Tedre et al., 2021; Wu, 2021). Given the large volume of primary studies on AI, more research attention has been directed towards scrutinizing AI's effects on education via systematic reviews. For example, a Google Scholar search by using two strings of "systematic review" and "AIED" led to about 1,980,000 results, which includes many AIED studies conducted in systematic review methodology. Although numerous systematic reviews exist, scarce tertiary reviews (e.g., Adadi et al., 2022; Bond et al., 2024), also known as reviews of systematic reviews, have synthesized the results of those secondary studies to offer a panoramic view of the field and identify limitations in existing literature to inform future research trajectory. Further, few tertiary reviews thoroughly examined AI's multifaceted impact in diverse educational contexts, including both PreK-12 education and higher education.

To fill this gap, we conducted a tertiary review to investigate the realm of AIED and highlight underexplored research areas based on our analysis. Specifically, we applied a scoping meta-review (SMR) and bibliometric analysis to examine systematic reviews on AIED. Following the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines (Page et al., 2021), we collected 126 systematic reviews from Scopus and Web of Science (WoS) for our study. By analyzing those reviews, our study generated four themes that describe the current state of AIED: (1) the key characteristics of the included systematic reviews, (2) AI impact on students' learning, (3) AI impact on educators' teaching, and (4) AI impact on institutional administration and service in education. We also compared and contrasted PreK-12 education and higher education and provided suggestions to improve future research.

Related Work

Despite the existence of numerous systematic reviews on AIED, very limited tertiary reviews have synthesized the results of those reviews to provide a holistic understanding on the effects of AI use in the broad educational landscape. Among the limited research, Adadi et al. (2022) employed a systematic umbrella review approach to examine the role of AI applications in COVID-19 and shed light on a post-pandemic research agenda. They followed the PRISMA guidelines and searched PubMed, Scopus, and other preprint sources to collect relevant reviews on AI's role in COVID pandemic published from December 1st, 2019, to June 1st, 2021. Based on analysis of 45 eligible reviews, they found that studies on AI applications in medical settings predominated the literature and identified

traditional machine learning and deep learning as the most widely used AI methods. Further, they reported that few studies employed meta-analysis.

Different from Adadi et al.'s (2022) foci on AI use in various settings during the pandemic and post-pandemic vision on AI research, Bond et al. (2024)'s meta systematic review examined AI solely in higher education. Their analysis of 66 included reviews between 2018 and July 2023 revealed that the majority were systematic reviews published by authors from North America through team collaborations. They also found that adaptive systems and personalization were predominantly employed in higher education compared to other AI applications. Although both studies mentioned above highlighted the importance of analyzing AI's effects, neither study took the synergy of an SMR and bibliometric analysis to study both PreK-12 schools and higher education. In view of this absence, the current study intends to fill this gap.

Methodology

We conducted a tertiary review to examine systematic reviews on AIED. Following the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines (Page et al., 2021), we collected 126 systematic reviews from Scopus and WoS for our study. Specifically, we used bibliometric analysis and reported on key features of the included reviews, such as keyword co-occurrence analysis and co-authorship analysis. We also employed SMR (Sarrami-Foroushani et al., 2015) to analyze only systematic reviews on AI use in educational contexts and generate essential themes from relevant review results. The following section describes the search strategy, screening process and criteria as well as methods of data analysis.

Search Strategy

We conducted a preliminary review and two rounds of search to identify appropriate articles from Scopus and WoS, two databases indexing peer-reviewed articles. Following Sarrami-Foroushani et al.'s (2015) suggestion to undertake an SMR, we started with a nonsystematic preliminary review to explore the literature related to AI use. We did not only find a plethora of studies on AI, but also noted the diversity and complexity of literature due to numerous publications from different disciplines and various subcategories of emerging technologies under the big umbrella of AI. To limit the size of citations, we decided to focus our search on AI use in educational contexts. For the first round of search in Scopus and WoS, we probed more recent systematic reviews published between January 1st, 2020, and March 30th, 2023, and used two search strings "AI or artificial intelligence" AND "education" that were applied to all fields to cast a wide net. Based on the results from the first round, we carefully examined the search terms used in the included review articles and refined the search strings for our second round. Specifically, we applied three search strings to all fields as listed in Table 1 below, identifying systematic reviews, various types of AI along with diverse educational settings which were not

Table 1 Search strings used in the second round

Operator	Dimension/Topic	Search Terms
AND	Artificial Intelligence	AI or artificial intelligence or machine learning or deep learning or chatbot or robotic or social robotics or intelligent systems or expert systems or machine intelligence or learning analytics or neural network or support vector machine or natural language processing or data mining or adapted learning or automated assessment or algorithm or personalized learning or prediction system or learner analytics
	Education	Education or pre-kindergarten or pre-primary or daycare or kindergarten or childcare or early childhood or preschool or primary or elementary or middle or junior or senior or high school or secondary or college or university or higher education or graduate studies or postgrad or undergrad or post-secondary education or tertiary or pretertiary or K-12 or adult education or vocational education
	Review Type	Systematic review

comprehensively included in the first round of search. Unlike the first round, we did not specify the period of publication for the second round. This enabled us to include systematic reviews published by Mid-August of 2023 to decipher evolutionary nuances of the AIED research field.

Screening Process, Criteria and Quality Assessment

We applied the PRISMA guidelines (Page et al., 2021) and inclusion and exclusion criteria in Table 2 to inform our screening process, which consisted of two rounds of searches in WoS and Scopus (see Fig. 1). Specifically, we initially identified 88 articles from Scopus and 43,413 articles from WoS in the first-round search and 123 articles from Scopus and 27,236 articles from WoS in the second-round search. Due to the large corpus of literature in WoS, we selected "review article" as the document type and "Educational Research" as the category to refine our search, which helped remove 43,327 articles from the first-round search and 26,849 articles from the second-round search in WoS. Then, we examined the title, key words, and abstract of 88 articles from Scopus and 86 articles from WoS collected in the first

Table 2 Inclusion and exclusion criteria

Inclusion Criteria	Exclusion Criteria
1. The article's full text is available in English.	1.The article's full text was not written in English.
2. The article is published in peer-reviewed journals.	2. The article was not published in a peer-reviewed journal.
3. The article is a systematic review.	3. Nonsystematic reviews were excluded from this study.
4. The article focuses on AI use in one or multiple education context(s).	4. The article did not have a focus on AI use in one or multiple educational context(s).

round and 123 articles from Scopus and 387 articles from WoS collected in the second round. This further scrutiny of those reviews led to the retrieval of 20 articles from Scopus and 24 articles from WoS in the first round and 20 articles from Scopus and 95 articles from WoS in the second round. Upon completion of searches, we exported those 159 articles identified from both rounds to Zotero and used the software's merge function to remove duplicates, resulting in a total of 134 systematic reviews. We conducted a full text review of those reviews and found eight articles did not have a clear focus on AI use in education. As a result, a final 126 systematic reviews were retained in this study.

In addition, we employed the ten criteria used in Bond et al.'s study (2024) to assess the quality of the 126 reviews. As shown in Table 3, the included studies did best in reporting research questions, aims or objectives (99.2%), the inclusion/exclusion criteria (91.2%) and publication years (96.8%), followed by providing the search string (60.3%), review limitations (46.0%), data extraction coding scheme (42.1%) and search coverage (41.3%). By contrast, only a small number of studies totally met the criteria of reporting inter-rater reliability (22.2%), providing sufficient details about individual review (18.3%), and undertaking quality assessment (11.9%).

Data Analysis

Through examination of the findings of 126 systematic reviews, we intended to map the field of AIED by conducting an SMR which takes the synergy of bibliometric analysis and thematic analysis. To respond to the first question on key characteristics of the included reviews, we analyzed keyword co-occurrence, co-authorship, geographic distribution of authorship, time of publication, size range, educational levels, subject areas, types of AI applications or algorithms, as well as learning environment and modalities. The thematic analysis (Boyatzis, 1998) empowered us to answer the other three research questions by generating three major themes related to teaching, learning, and administration and further identifying sub-themes within each. As suggested by Sarrami-Foroushani et al. (2015), during the stage of results analysis and reporting, an SMR should "thematically analyze the selected texts" and "create a narrative to summarize the results" (p.78). Given that the major advantage of an SMR is "to map a new field based on

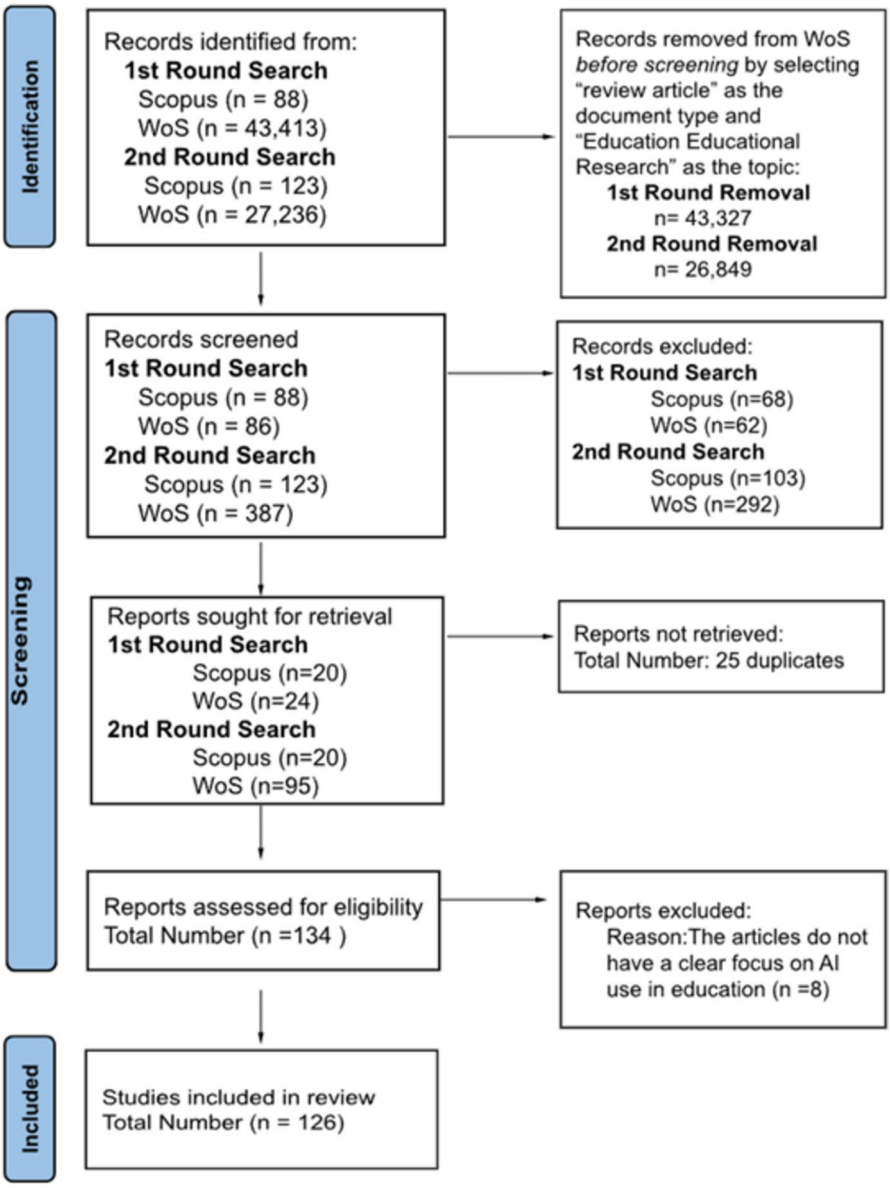

Fig. 1 PRISMA flow chart
This chart was adapted based on Page et al.'s (2021) PRISMA 2020 statement

high-level evidence provided by systematic reviews" (Sarrami-Foroushani et al., 2015, p. 81), we performed thematic analysis as a "process for encoding qualitative information" to generate themes inductively (Boyatzis, 1998, p. 4). Adopting a bottom-up approach, we developed the codes by reading through the included reviews.

Table 3 Quality assessment of the 126 reviews per Bond et al.'s (2024) modified criteria

Criteria	Totally met	Partially met	Not met	Not applied
1. Are there any research questions, aims or objectives?	99.2%	0.0%	0.8%	0.0%
2. Were inclusion/exclusion criteria reported in the review and are they appropriate?	91.2%	4.0%	4.8%	0.0%
3. Are the publication years included defined?	96.8%	1.6%	1.6%	0.0%
4. Was the search adequately conducted and likely to have covered all relevant studies?	41.3%	23.8%	34.9%	0.0%
5. Was the search string provided in full?	60.3%	37.3%	2.4%	0.0%
6. Do they report inter-rater reliability?	22.2%	9.5%	61.9%	6.4%*
7. Was the data extraction coding scheme provided?	42.1%	46.0%	11.1%	0.0%
8. Was a quality assessment undertaken?	11.9%	0.0%	81.8%	6.4%*
9. Are sufficient details provided about the individual included studies?	18.3%	73.8%	1.6%	6.4%*
10. Is there a reflection on review limitations?	46.0%	25.4%	28.6%	0.0%

Reviews adopted bibliometric analysis and text-mining approaches did not apply to criteria 6,8, and 9, which were marked * in the table

To increase reliability, we carefully read the systematic reviews from the first round of search and independently coded the selected texts to generate themes inductively from the data. We also constantly compared our codes of those reviews during meetings and created a codebook that summarized the key categories of our codes by team consensus. Guided by those key categories, we independently coded the rest of systematic reviews collected from the second round of search and updated the codebook based on our analysis of those reviews. The inter-rater agreement percentage was 91.6%. Upon finishing coding of all reviews, we held meetings to discuss and finalize themes and sub-themes revealed from our coding process, which is shown in Fig. 10 in the following section.

Findings

Key Characteristics of the Included Studies

We conducted keyword co-occurrence analysis and co-authorship analysis to understand and visualize the internal structure of the included reviews and relationship among authors. We also delved into each systematic review and summarized some key characteristics in Appendix A, such as review size and publication years, country of the first author, AI technologies, educational levels, subjects, learning environments, and modalities.

To begin with keyword co-occurrence analysis, a total of 678 keywords appeared in the 126 included reviews; and 37 keywords met the threshold with a frequency of at least 5 appearances. The five keywords with the highest occurrences are systematic review (61), artificial intelligence (55), machine learning (25), online learning (24), education (24) while keywords with strongest total link strength are systematic review (214), artificial intelligence (135), online learning (106), learning analytics (90), and performance (86). Figure 2 shows that the relationship among the 37 keywords can be grouped into five clusters as represented by different colors; and there are intricate interconnections between different clusters. In specific, a cluster is a set of closely related keywords/nodes that represent research areas. We also looked into those five clusters and presented relevant information about each cluster in Table 4. Among the five clusters, cluster one in red and cluster two in green have the largest number of keywords. In particular, we found that online learning, learning analytics, performance, higher education, model, and data mining in cluster one and artificial intelligence, framework, impact, knowledge, trends in cluster two have both high occurrences and total link strength, indicating those topics attracted much research attention.

The co-authorship analysis revealed that 20 out of the 430 authors contributed to a minimum of two systematic reviews. In detail, we found Hwang to be the most productive author who completed four reviews; and for the other 19 authors, each completed two reviews. Figure 3 shows the collaborative network of those 20 authors. Though many authors did not have a cooperation relationship with others, we observed that the largest set of connections shown in the middle of the figure involved six authors, including international collaboration between Pishtari and

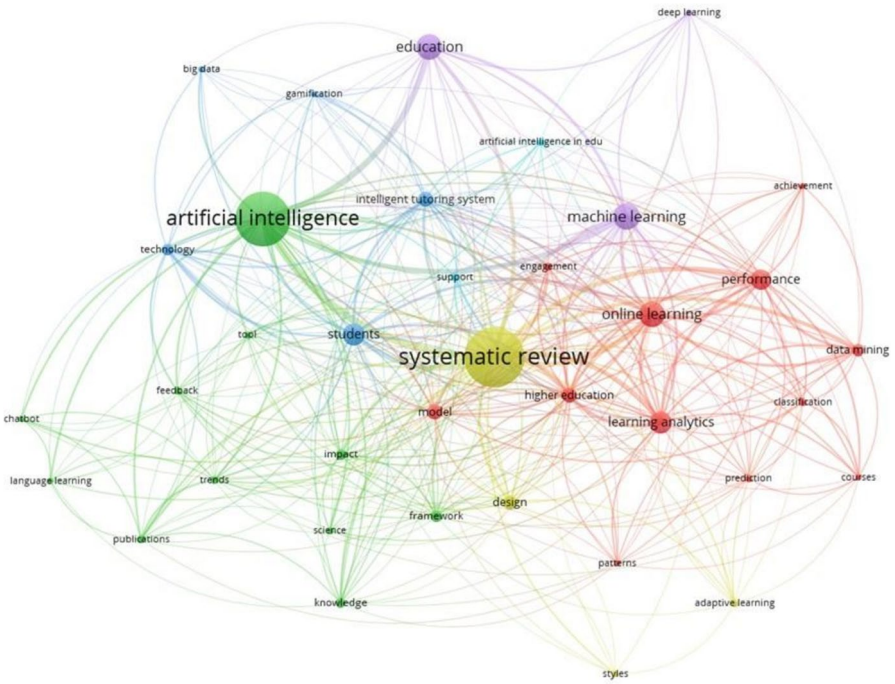

Fig. 2 The keyword co-occurrence network of the included reviews (using VOSviewer)

Serrano-Iglesias (i.e., the cluster in yellow) and domestic collaboration among Serrano-Iglesias, Alario-Hoyos, Martinez-Mones, Kloos, and Dimitriadis (i.e., the cluster in red). Additionally, there are three small sets of connections depicted in the figure, including domestic collaboration between Wibowo and Zhai (i.e., the cluster in purple), among Tu, Hwang, and Chang (i.e., the cluster in green), and among Mathrani, Susnjak and Umer (i.e., the cluster in blue).

In respect of geographic distribution of authorship (Fig. 4), we analyzed the first author's country/region in the included reviews and found that most studies were conducted in Asia (49%), followed by Europe (25%). Far less systematic reviews were conducted in North America (9%), Oceania (7%), Africa (6%), and South America (4%). Furthermore, the authorship was distributed across 40 different countries/regions, with China, Spain, and the United States as the top three countries that contributed the greatest number of studies in this review.

In terms of the numbers of reviews published throughout the time (see Fig. 5), it was interesting to note that even though the earliest review was published in 2014, there were none published from 2015 to 2018 included in this study. Since 2019, there has been a steady increase in the number of systematic reviews published in each year, with 7 studies in 2019, 11 in 2020, 23 in 2021, 39 in 2022, and 46 in 2023 up to mid-August.

Though the sizes of the included reviews varied considerably, the majority of the reviews examined less than 100 studies. As shown in Figs. 6 and 23 reviews had a

Table 4 The keyword information for five clusters

Cluster	NO. of Keywords	Keywords	Occurrences	Total Link Strength
1 (Red)	12	online learning	24	106
		learning analytics	20	90
		performance	18	86
		higher education	13	74
		model	13	56
		data mining	11	50
		engagement	7	39
		classification	7	30
		prediction	6	28
		courses	5	27
		patterns	5	23
		achievement	5	22
2 (Green)	11	artificial intelligence	55	135
		framework	10	38
		impact	9	45
		knowledge	8	33
		trends	7	36
		publications	7	32
		feedback	6	32
		tool	6	27
		science	6	23
		language learning	5	18
		chatbot	5	17
3 (Blue)	7	students	20	84
		intelligent tutoring system	13	58
		technology	10	49
		support	7	33
		artificial intelligence in education	6	24
		gamification	5	25
		big data	5	15
4 (Yellow)	4	systematic review	61	214
		design	12	50
		adaptive learning	6	18
		styles	5	10
5 (Purple)	3	machine learning	25	85
		education	24	62
		deep learning	5	20

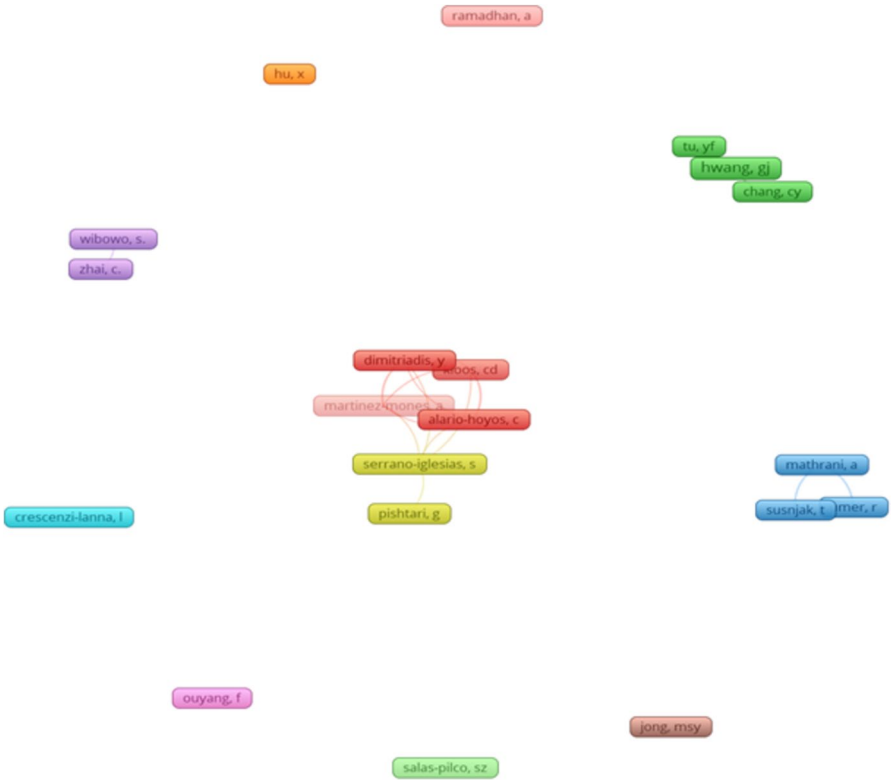

Fig. 3 The co-authorship network of the included reviews (using VOSviewer)

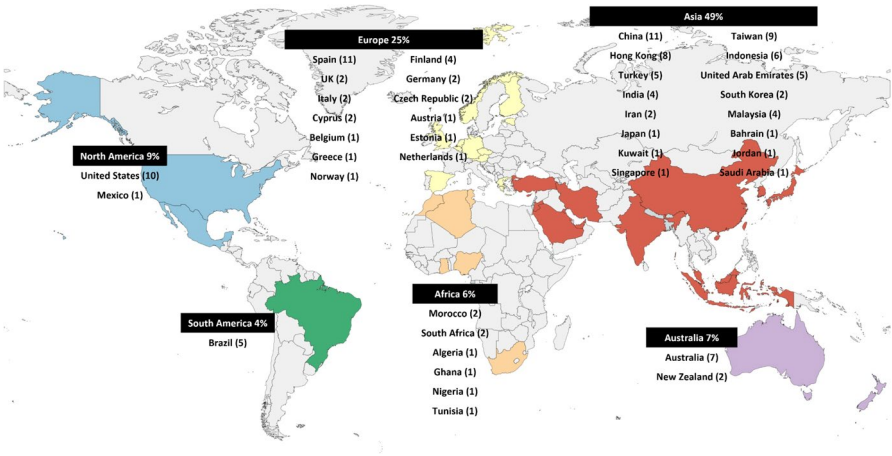

Fig. 4 Geographic distribution of authorship (using mapchart.net)

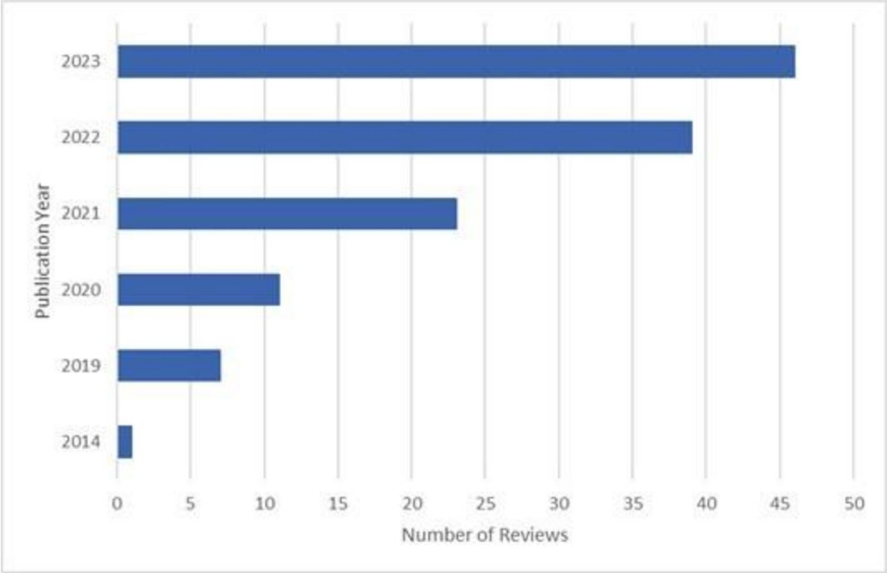

Fig. 5 Time period of publication

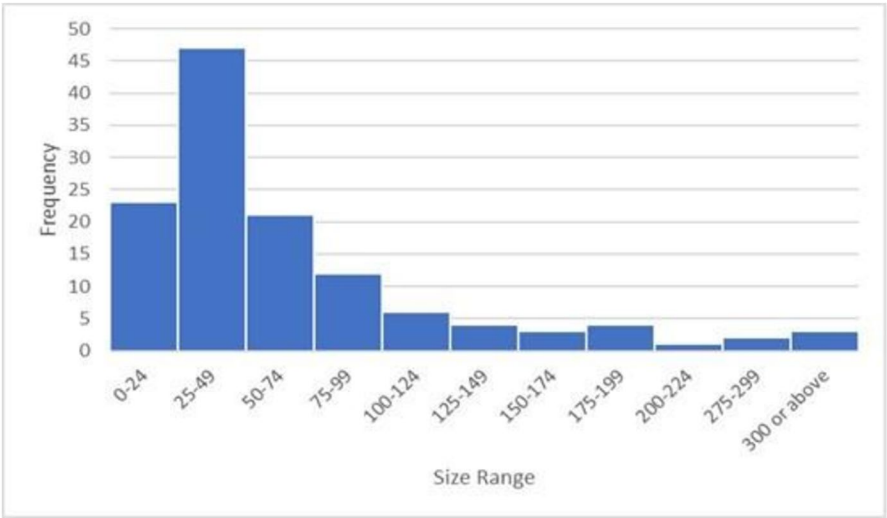

Fig. 6 Size range of the included reviews

size range of less than 24 studies, 47 reviews a size range of 25–49, 21 reviews a size range of 50–74, and 12 reviews a size range of 75–99. By contrast, far less reviews had a size range above 100 studies, including a total of 17 studies with a size range of 100–199, 3 studies with a size range of 200–299, and 3 studies with a size range of 300 or above. Among those six studies with a size range above 200, four studies

(i.e., Ahadi et al., 2022; Bozkurt et al., 2021; Dogan et al., 2023; Feng & Law, 2021) employed text-mining approach and/or bibliometric analysis. By contrast, only two (i.e., Pradana et al., 2023; Tang et al., 2023) out of the 120 studies, which have a size range less than 200, used bibliometric analysis. While most of the included reviews used qualitative analysis (e.g., content analysis or thematic analysis), reviews with a large size range above 200 were more likely to extract structured information via text-mining or identify trends and patterns via bibliometrics.

To scrutinize the diverse educational settings involved in the included reviews, we analyzed educational levels, subject areas, learning environment, and learning modalities. In respect of educational levels (Figs. 7), 42% of the reviews centered on general AIED without providing detailed information about the type of education examined in individual studies. Among those reviews that specified educational levels, 31% were focused on multiple educational levels that covered both higher education and PreK-12 education, 16% on higher education, and 11% on PreK-12 education. It is worth mentioning that 20 out of the 39 reviews on multiple education levels reported that most included studies were conducted in higher education, evidencing how higher education learning contexts predominate AIED research.

Regarding the subjects (Fig. 8), almost half of the included review did not specify the learning content areas. About 29% of the reviews investigated studies conducted in a wide variety of subject areas with computer science, science, engineering, and mathematics as the most popular subjects. In addition, 22% of the reviews were focused on a single subject area with language education as the most studied; and other single subject area examined in the reviews included chemical education, mathematics education, nursing education, music education, special education, teacher education, health education, radiology and medical education, and AI education.

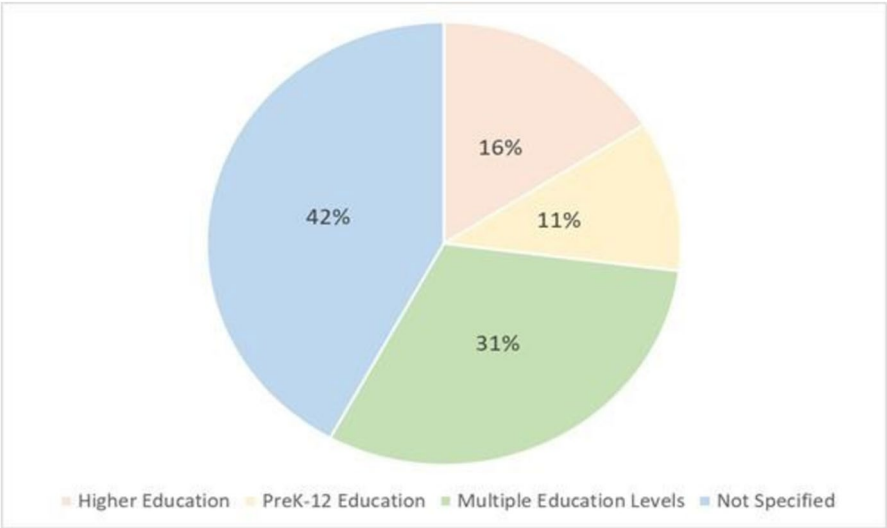

Fig. 7 Educational levels

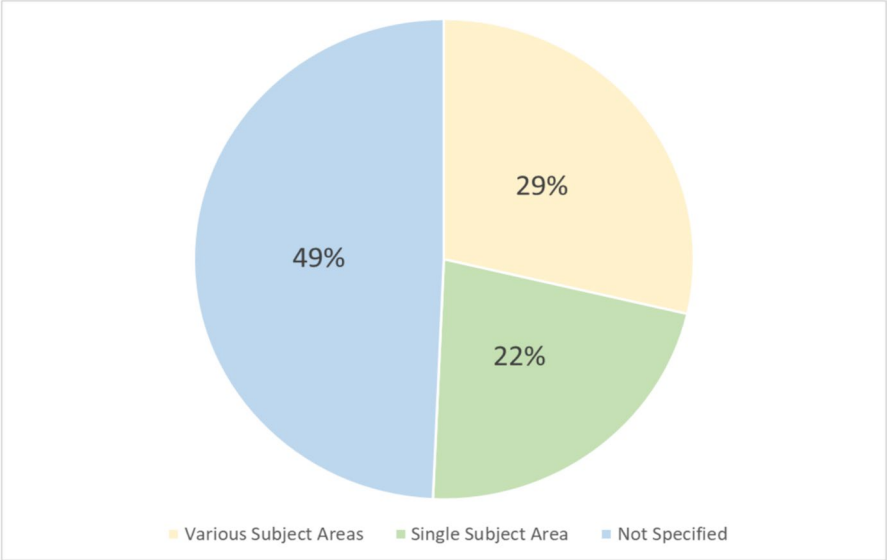

Fig. 8 Subject areas

Out of the 126 reviews, 110 studies did not provide information about the learning environment. In detail, only six included reviews addressed on formal classroom learning, one on experimental informal learning, nine on both formal and informal learning environments that are within and outside of the regular classrooms (e.g., research laboratory, workshops, and after-school lessons). Similarly, 83 out of 126 reviews did not mention learning modalities examined in the included studies. The most widely reported modality is online learning, taking up about 28% of the included reviews while none of the reviews was solely focused on face-to-face learning, one review on blended learning, and seven reviews on multiple learning modalities.

Apart from examining the diverse learning contexts, we also analyzed types of AI applications or algorithms employed in the individual reviews. As visualized in Fig. 9 that is based on word frequency counts, the most widely studied AI was machine learning (43 appearances), followed by intelligent tutoring systems (17 appearances), learning analytics (18 appearances), and data mining (14 appearances).

Themes and Subthemes of AIED

Our thematic analysis revealed three major themes and nine subthemes related to AI's impact on learning, teaching, and administration. Figure 10 below presents a thematic map of AIED that captures the relationships between those themes and subthemes. Guided by this map, we unfolded each theme in the following section. One clarification is selected reviews on AIED emphasize more on the strengths

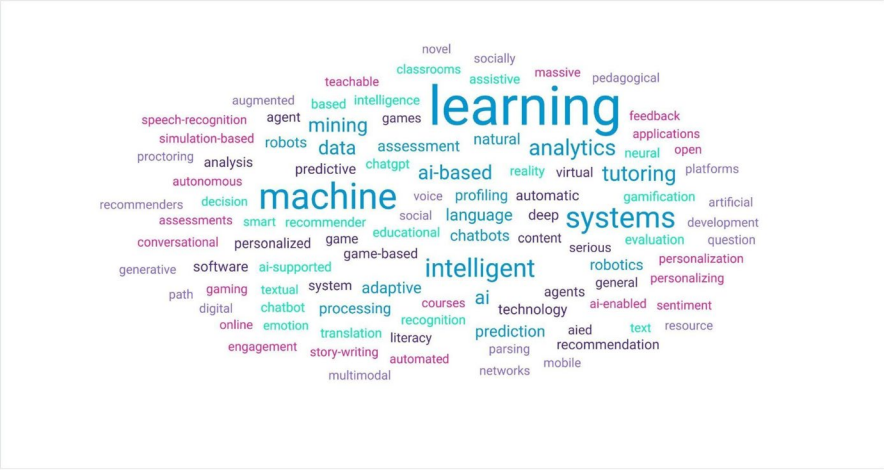

Fig. 9 AI applications/algorithms (using Word Cloud Generator)

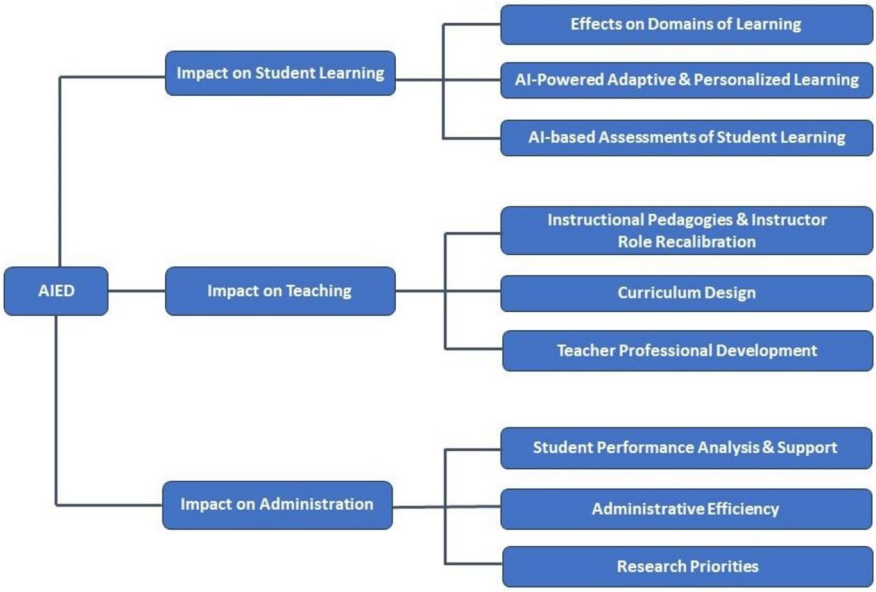

Fig. 10 Thematic map of AIED

of AI use in education. Limited reviews (e.g., Martinez-Roig et al., 2023; Papadopoulos et al., 2020; Ramadhan et al., 2023; Santomartino & Yi, 2022; Sikstrom et al., 2022; Sun et al., 2023; Umer et al., 2023; Wang et al., 2023; Yan et al., 2023; Zawacki-Richter et al., 2019) briefly mentioned the negative or neutral impact of AI use. For example, Martinez-Roig et al. (2023) revealed that adding humor to robot's dialogues negatively influenced children's confidence. Ramadhan et al. (2023)

showed the negative impact of intelligent tutor systems with gamification combination, which could divert students' attention from educational goals. In another study, Yan et al. (2023) discussed the potential risk of adopting underperforming AI models that could negatively affect students' learning experiences.

AI's Impact on Student Learning

AI has the potential to significantly impact student learning in various ways. Based on the thematic review of the selected articles, themes, including effects on domains of learning, adaptive and personalized learning, and AI-based assessments were identified.

Effects on Domains of Learning First and foremost, the effectiveness of AI in influencing various domains of student learning, encompassing cognition, affect, 21st-century skills and learning strategies, has undergone extensive exploration. The impact of AI on cognition, specifically in terms of learning performance, higher-order thinking skills, and collaboration or communication, has been the subject of investigation in recent studies (e.g., Huang et al., 2022; Moore et al., 2023; Ng et al., 2023; Sikstrom et al., 2022; Wang et al., 2023; Yue et al., 2022). The key objective in most studies was to enhance learning outcomes and facilitate interaction between human and AI technologies. To illustrate, Moore et al. (2023) conducted a review encompassing 12 studies published between 2014 and 2021, aiming to unveil the instructional contexts and applications of artificial intelligence in K-12 learning environments. The systematic review underscores the promising potential of AI in supporting student learning.

Concerning affect, it encompasses attitudes, self-efficacy or confidence, satisfaction or interest, and emotion. This aspect has been explored in eight studies (Ba & Hu, 2023; Chu, Hwang, & Tu, 2022; Fang et al., 2023; Huang et al., 2022; Liang et al., 2021; Rice & Dunn, 2023; Rizvi et al., 2023; Sikstrom et al., 2022). AI systems can be designed to recognize and respond to students' emotions. Moreover, AI offers motivational support by setting achievable goals, providing positive reinforcement, and offering personalized challenges. AI-powered systems can also deliver timely and constructive feedback, thereby contributing to a positive learning environment. Positive reinforcement and encouragement, as observed in Rice and Dunn (2023), can significantly boost students' confidence and enthusiasm for learning. This systematic review focused on assistive technology for students with identified disabilities and found that AI played a contributory role in enhancing students' emotional well-being and enjoyment of learning when utilizing AI software.

In addition to the effectiveness of AI on cognition and affect, seven studies (Blumenstein, 2020; Liang et al., 2021; Sikstrom et al., 2022; Xu & Ouyang, 2022b; Yang & Kyun, 2022; Younis et al., 2023; Zhai & Wibowo, 2022) have explored its influence on students' 21st-century skills, including creativity (Younis et al., 2023), critical thinking (Younis et al., 2023), communication (Blumenstein, 2020; Liang et al., 2021), and problem-solving skills (Sikstrom et al., 2022; Xu & Ouyang, 2022b) or learning strategies, such as higher-level cognitive skills (Liang et al., 2021; Yang & Kyun, 2022b; Zhai & Wibowo, 2022) and metacognition

(Blumenstein, 2020; Liang et al., 2021; Sikstrom et al., 2022). Metacognition, for example, includes self-regulation, learning to learn, information-seeking, monitoring, and modifying attitudes and behaviors toward certain learning goals. It has been reviewed that AI can support metacognitive skills by prompting students to reflect on their learning experiences (e.g., Sikstrom et al., 2022). This support includes setting goals, monitoring progress, and adjusting learning strategies. This area of AI impact is closely related to the adaptive and personalized learning discussed further in the next section.

AI-Powered Adaptive and Personalized Learning Thirteen systematic reviews (Bhutoria, 2022; Blumenstein, 2020; Bozkurt et al., 2021; Daoudi, 2022; Durso & Arruda, 2022; Huang et al., 2022; Hwang & Tu, 2021; Khakpour & Colomo-Palacios, 2021; Salas-Pilco, Xiao, & Oshima, 2022; Santos et al., 2022; Tabuenca et al., 2021; Weng & Chiu, 2023; Zhai et al., 2021) shed light on how AI can adapt and personalize learning for students. Specifically, AI-powered adaptive learning systems can dynamically adjust the difficulty level of content based on a student's performance. This ensures that students are appropriately challenged and engaged, contributing to a more effective learning experience. This adaptiveness inherent in AI is closely associated with personalized learning. AI has the capability to analyze an individual student's learning styles, strengths, and weaknesses, tailoring educational content accordingly. This personalized approach facilitates students in learning at their own pace, addressing their specific needs, and fostering a deeper understanding of the subject matter.

The overarching goal of AI-powered adaptive and personalized learning is to optimize the learning process by delivering customized content, adapting to individual pacing, and offering personalized feedback. This approach aims to create an inclusive platform catering to all types of learners, ultimately enhancing the overall effectiveness of education. For instance, Hwang and Tu (2021), through a bibliometric mapping analysis and systematic review focused on exploring the role and research trends of AI in mathematics education, discovered that the role of AI in "adaptive systems and personalization" (p. 584) lies in tailoring guidance or support to students at various proficiency levels by analyzing their learning status and behaviors.

AI-based Assessments of Student Learning The integration of AI in student assessment not only streamlines administrative tasks but also enhances the quality and personalization of the evaluation process. One noteworthy application of AI in education is automation. According to thirteen reviews (Alenezi & Faisal, 2020; Alonso-Fernandez et al., 2019; Celik et al., 2022; Cuellar-Rojas et al., 2022; Gonzalez-Calatayud et al., 2021; Kaddoura et al., 2022; Martínez-Comesaña et al., 2023; Okewu et al., 2021; Ouyang et al., 2022; Shin & Shim, 2021; Xu & Ouyang, 2022b; Yang & Kyun, 2022; Zawacki-Richter et al., 2019), AI-supported automatic assessment not only evaluates students' performances and provides immediate feedback,

but also increases the accuracy and objectivity of grading or assessment, ensuring the fairness and reliability of student scores.

For instance, Ouyang et al. (2022) offer an overview of empirical research on AI applications in online higher education. The results indicate that automatic assessment stands out as a primary function of AI in online higher education. AI has been effectively utilized to automatically assess students' performances and learning capacities, deliver timely feedback, and enhance students' self-awareness and self-reflection. Another study by Martínez-Comesaña et al. (2023), synthesizing nine research endeavors focusing on improvements in primary/secondary student assessment using AI tools, highlights a significant contribution. The application of AI in assessing students' performance automates evaluations, making them more objective and fairer through the implementation of neural networks or natural language processing.

AI's Impact on Teaching

AI is reshaping instructional pedagogies, transforming curriculum design, and revolutionizing teacher professional development. As the education landscape continues to evolve, the thoughtful integration of AI technologies holds the potential to improve the overall quality and effectiveness of teaching and learning.

Instructional pedagogies and instructor role recalibration The influence of AI on instructional pedagogies is transformative, necessitating a reassessment of instructor roles. Eleven studies, including those by Crompton et al. (2022), Fariani et al. (2023), Ley et al. (2023), Ng et al. (2023), Pishtari et al. (2020), Rizvi et al. (2023), Sanusi et al. (2023), Tabuenca et al. (2021), Yue et al. (2022), Zhai and Wibowo (2023), and Zhu et al. (2022), have examined the advantages of incorporating AI into specific pedagogical approaches such as active learning, model-based learning, personalized learning, gaming, and collaborative project-based learning. These studies frequently employ various terms to refer to virtual instructors with similar meanings, including intelligent tutor, cognitive tutor, animated pedagogical agents, and intelligent avatars (Crompton et al., 2022). Taking Sanusi et al. (2023) as an illustrative example, a systematic review of 43 sources delved into the landscape of machine learning (ML) teaching and learning in K-12 settings. The review disclosed that ML has been integrated across K-12 levels, from primary to high school. In teaching ML, diverse pedagogical approaches were explored, with the most prevalent approach involving project-based learning. This approach emphasizes collaborative efforts in co-creating ML-based solutions with students.

The integration of AI into teaching not only transforms the learning experience for students but also necessitates a shift in the roles and responsibilities of educators, placing them at the forefront of guiding students in the AI era. First, teachers transition from traditional roles to facilitators of personalized learning experiences. Five reviews (Bhutoria, 2022; Gentile et al., 2023; Ji et al., 2023; Kang, 2023; Kuhail

et al., 2023) indicate that AI enables the customization of content and learning paths based on individual student needs, requiring teachers to guide and support students on more individualized journeys. Second, teachers become curators of educational content. With AI assisting in content delivery, teachers play a crucial role in selecting, organizing, and contextualizing AI-generated resources to suit the specific needs and goals of their students (e.g., Celik et al., 2022). Third, teachers evolve into data-driven instructors. AI provides valuable insights into student performance and learning patterns, empowering teachers to make informed decisions about instructional strategies, interventions, and resource allocation (e.g., Chu, Hwang, Tu, & Yang, 2022). Also, teachers take on the role of mentors in fostering AI literacy among students. They guide students in understanding how AI works, its applications, and the ethical considerations associated with its use, preparing students for the AI-driven future (e.g., Ng et al., 2023; Sanusi et al., 2023). Further, teachers adopt innovative pedagogies supported by AI. This includes flipped classrooms, gamified learning experiences, and project-based approaches that leverage AI tools to enhance student engagement and learning outcomes (e.g., Rizvi et al., 2023).

Curriculum design AI has the potential to exert a significant influence on the design of educational curricula, particularly through the incorporation of AI in teaching methodologies and the introduction of AI literacy courses. It has been observed that AI can effectively analyze extensive educational data to discern trends, patterns, and areas requiring curriculum enhancements (Blumenstein, 2020). Curriculum designers leverage this information to make informed, data-driven decisions, refining and updating content to better cater to student needs and align with evolving educational landscapes (e.g., Blumenstein, 2020). Concerning AI literacy, the inclusion of AI as a subject is recommended in K-12 education, either as an integrated part of the curriculum or through extracurricular activities, possibly condensed into short-term courses spanning approximately 15 h. As highlighted by Yue et al. (2022), the majority of AI education teaching units predominantly involve in-person and extracurricular activities conducted in lab settings, workshops, or after-school sessions. Notably, AI education is still not a standard component in regular classrooms, including those dedicated to technology and computing-related courses. The prevalence of short-term courses suggests that a brief yet focused lesson may be adequate for children to grasp fundamental AI concepts. Furthermore, AI literacy courses ought to encompass essential AI principles, applications, and/or ethical considerations and security measures (Casal-Otero et al., 2023; Sanusi et al., 2023). Therefore, it is inferred that the design of AI literacy curricula should adopt an interdisciplinary approach, incorporating elements from computer science, mathematics, ethics, sociology, and other pertinent fields. This interdisciplinary strategy empowers students to comprehend the multifaceted impact of AI across various disciplines, fostering a holistic perspective.

Teacher professional development Most studies on AI focus on pedagogical development with a dearth of teacher professional training and development (Sanusi et al., 2023). Based on Casal-Otero et al.'s (2023) systematic review, the integration of AI in teaching and learning has a substantial impact on teacher professional

development, influencing how teachers acquire new skills, adapt to evolving technologies, and enhance their teaching practices. For example, teachers develop AI literacy, gaining a deeper understanding of how AI works, its applications in education, and its potential impact on teaching and learning (e.g., Casal-Otero et al., 2023). This literacy equips teachers to make informed decisions regarding AI integration. Further, professional development becomes a continuous and adaptive process. Teachers engage in ongoing learning experiences to stay abreast of new AI technologies, emerging pedagogical approaches, and best practices for integrating AI into the classroom (e.g., Gentile et al., 2023). In addition, teachers participate in AI-assisted training programs that leverage AI technologies to identify areas for improvement (e.g., Ng et al., 2023; Sanusi et al., 2023). These programs provide targeted resources, feedback, and learning materials tailored to individual needs.

AI's Impact on Administrative and Institutional Service

AI holds the potential to profoundly influence administrative and institutional services across a spectrum of areas. These encompass, among others, the anticipation of student academic achievement and assistance delivery, enhancements in administrative efficiency, and the discernment of research priorities.

Student performance analysis and support Higher education grapples with a significant challenge, which is the notably high student dropout rates, particularly in undergraduate programs (Mercado, 2024). In response to this issue, universities have embarked on reforms to mitigate dropout rates. In this context, the integration of AI applications can play a pivotal role in predicting student attrition, uncovering the determinants influencing their academic performance, and delivering essential support mechanisms. Sixteen reviews (e.g., Abu Saa et al., 2019; Chiu, 2021; Chu, Hwang, Tu, & Yang, 2022; Costa et al., 2020; Crescenzi-Lanna, 2020; Crompton et al., 2022; Daoudi, 2022; Durso & Arruda, 2022; Grimalt-Alvaro & Usart, 2023; Khan & Ghosh, 2021; Moreno-Marcos et al., 2018; Papamitsiou & Economides, 2014; Ramaswami et al., 2023; Sekeroglu et al., 2021; Umer et al., 2023; Yan et al., 2023) have identified how AI can be used to predict students' academic performance and nine reviews (Agrusti et al., 2019; Albreiki et al., 2021; Bin Roslan & Chen, 2022; Chen et al., 2022; Costa et al., 2020; Ifenthaler & Yau, 2020; Kaddoura et al., 2022; Salas-Pilco & Yang, 2022; Sghir et al., 2023) reveal how AI can be used to anticipate student dropout rates. Furthermore, AI solutions have been explored to provide the necessary support systems (e.g., Banihashem et al., 2022; Crescenzi-Lanna, 2020; Okonkwo & Ade-Ibijola, 2021).

For instance, in the study by Okonkwo and Ade-Ibijola (2021), they highlighted the effectiveness of a conversational agent in supporting students with administrative assistance, including guidance on matters such as orientation, enrollment, and retention. The research also demonstrated that Chatbots can carry out diverse administrative tasks within the educational process, such as grading assignments and offering feedback to students. The utilization of Chatbots

for administrative functions streamlines students' access to critical information, such as admission processes, scholarships, and tuition fees. In essence, AI serves as a valuable tool for educational stakeholders, aiding them in identifying and addressing specific areas where students encounter academic challenges. This, in turn, promotes efforts geared toward enhancing students' educational achievements and elevating the overall quality of education.

Administrative efficiency AI has a profound influence on enhancing administrative efficiency, manifesting in various domains, including the automation of repetitive tasks, data analysis and decision support (e.g., Salas-Pilco & Yang, 2022), the integration of AI-powered chatbots and virtual assistants (e.g., Okonkwo & Ade-Ibijola, 2021), efficient document management (e.g., Chiu, 2021), personalized services (e.g., Chiu, 2021; Umer et al., 2023), resource allocation (Kaddoura et al., 2022; Nigam et al., 2021; Zawacki-Richter et al., 2019), and cost reduction (e.g., Okonkwo & Ade-Ibijola, 2021; Salas-Pilco & Yang, 2022). For instance, Salas-Pilco and Yang's (2022) study shows AI prediction models proficient in estimating educational processes by leveraging extensive educational datasets. Predictive modeling is particularly advantageous as it offers precise insights into educational inquiries, enabling proactive decision-making while saving both time and resources. Additionally, intelligent analytics, powered by AI algorithms, adeptly processes and preprocesses abundant educational data for statistical analysis, providing compelling evidence of classification outcomes. Intelligent analytics are also invaluable in evaluating academic research performance and scientific productivity, facilitating a deeper understanding of and optimization of higher education systems through a classification process. In a separate investigation, Okonkwo and Ade-Ibijola's (2021) research demonstrates the utility of chatbots in executing administrative tasks within educational institutions. Their findings highlight the deployment of a chatbot system that automatically responds to student inquiries regarding the delivery of education. This digital assistant, in its capacity, offers guidance to tertiary institutions on enhancing their existing services, fostering novel innovative concepts, and effectively reducing labor costs.

Research Priorities While there has been limited exploration of how AI can enhance research prioritization, Salas-Pilco and Yang's (2022) review have shed light on the application of AI in research to streamline administrative tasks. The review highlights that higher education institutions are generally committed to enhancing the quality of their services, which directly impacts their responsibilities in teaching and research, ultimately affecting the university's efficiency and performance. Interestingly, AI methodologies offer a valuable tool to assess university performance and facilitate strategic decision-making by efficiently gathering essential data given that university performance assessment and strategic decision-making are key domains of higher education administration. Furthermore, research communities contribute significantly to a country's progress, making AI's role in identifying scientific communities and prominent research domains pivotal for policymakers seeking to establish research priorities.

Discussion and Implications

The current study expanded understanding about AI use in a variety of learning contexts. In the following section, we discussed key features of 126 systematic reviews as well as commonalities and nuanced differences between PreK-12 and higher education in terms of AI's impact on learning, teaching, and administration. Based on our findings, we provided implications for future studies.

Key Features of Included Reviews

It is interesting to note that the keyword co-occurrence network and analysis of AI applications in individual reviews both showed machine learning as the most frequently studied AI type, followed by intelligent tutoring systems, learning analytics, and data mining. A further look at the keyword co-occurrence network revealed how machine learning was related to various technologies, consisting of big data, gamification, deep learning, intelligent tutoring systems, data mining, and learning analytics. These findings support that machine learning plays a dominant role in the field of AIED apart from its wide use in medical settings during the COVID-19 pandemic (Adadi et al., 2022).

Additionally, our analysis of co-authorship confirmed Bond et al.'s (2024) finding that most research collaboration was domestic, and there was less international collaboration between authors. In terms of geographic distribution of authorship, we found more systematic reviews were conducted in Asia and Europe than other continents, which is different from North America's dominance in conducting secondary reviews in higher education (Bond et al., 2024). Moreover, China, Spain, and the United States were found as the most productive countries despite the widespread authorship across 40 different countries/regions.

By examining the time of publication, we noticed a steady increase of systematic reviews since 2019, like the trend reported by Adadi et al. (2022) and Bond et al. (2024). The rising number of reviews could be propelled by the growing interest in applying and researching AI to tackle COVID-19 crisis and challenges in post-pandemic landscape (Abrenilla et al., 2023; Dwivedi et al., 2023; Khan et al., 2021; Vaishya et al., 2020).

Although most included reviews had a study size below 100 and mainly employed qualitative analysis, we found that reviews (e.g., Ahadi et al., 2022; Bozkurt et al., 2021; Dogan et al., 2023; Feng & Law, 2021) that analyzed more than 200 studies were more likely to use text-mining approach and/or bibliometric analysis as the methodology. This might be explained by the advantages of applying text-mining and bibliometrics within systematic reviews. For example, use of text-mining can help reduce the time to complete research, and facilitate literature identification, category descriptions, and summarization (Thomas et al., 2011). By comparison, conducting bibliometric analysis enables researchers to rigorously decode and depict a large quantity of unstructured data (Donthu et al., 2021). Coupling with their strength in efficiently making sense of data, both approaches could be employed to

visualize or map a given research topic via interface-based or command-based softwares (Donthu et al., 2021; Linnenluecke et al., 2020).

We also found that many systematic reviews contained insufficient information to unpack the educational contexts, such as educational levels, subjects, learning environment, modality, and student characteristics. For instance, most studies did not specify the educational level(s) or intermingled PreK-12 education and higher education in the same review; and about half of the included studies did not report the subjects involved. A closer look at the educational level revealed that more studies were conducted in higher education than PreK-12 education. Within reviews that examined various subjects, computer science, science, engineering, and mathematics were widely studied (e.g., Alonso-Fernandez et al., 2019; Blumenstein, 2020; Celik et al., 2022; Chu et al., 2022). For reviews centering around a single subject, however, language education has drawn the most attention of researchers (e.g., Jeon et al., 2023; Ji et al., 2023; Liang et al., 2021), which reinforced Chalmers et al.'s (2023) finding about exponential increase in language education systematic reviews. Moreover, investigations of the learning environment (e.g., formal learning in regular classrooms and informal learning in labs, museums or at home) and modalities (e.g., online, blended, face-to-face and multiple modalities) were largely neglected in many of the included reviews. It is worth mentioning that only two out of the 126 reviews (Hopcan et al., 2022; Salas-Pilco et al., 2022) focused on AI's influence on diverse student populations (i.e., students with disability in special education contexts and minority students in inclusive education contexts) while the rest of the reviews were mainly structured around various AI technologies and rarely reported the demographic characteristics of the participating students. The incomplete information about these different aspects of learning may inhibit scholars and educators from developing an in-depth contextualized understanding of AI's effectiveness, which reiterates the importance in granular analysis of diverse and authentic contexts where AI has been applied (Eguchi et al., 2021; Hwang et al., 2023; Liao et al., 2022; Oyelere et al., 2022).

Commonalities and Nuanced Differences

Apart from scrutinizing those key features, we compared and contrasted findings related to PreK-12 education and higher education in light of student learning, teaching, and administrative and institutional service. To begin, our synthesis supported the powerful influence of AI on learning domains, adaptive and personalized learning, and assessments. By comparing PreK-12 education and higher education, we found a commonality that both educational settings extensively examined the impact of AI on cognition and affect (e.g., Ba & Hu, 2023; Huang et al., 2022; Moore et al., 2023; Rice & Dunn, 2023; Rizvi et al., 2023; Yue et al., 2022). On the contrary, fewer studies investigated how AI may influence students' 21st century skills, higher level cognitive skills or metacognition; and they were mostly pertaining to higher education (e.g., Blumenstein, 2020; Liang et al., 2021; Yang & Kyun, 2022). Additionally, reviews related to adaptive and personalized learning (e.g., Durso & Arruda, 2022; Zhai et al., 2021) and AI-based assessments (e.g., Okewu et al., 2021;

Ouyang et al., 2022; Zawacki-Richter et al., 2019) were mostly conducted in higher education or multiple educational levels with dominance in higher education. This confirmed Bond et al's (2024) finding about adaptive and personalized learning which has been predominantly employed in higher education compared to other AI applications.

In comparison to the overall amount of research examining AI's impact on learning, less reviews focused on its influence on teaching. One similarity between PreK-12 education and higher education was investigating how AI has been transforming instructional pedagogies and recalibrating instructors' roles (e.g., Crompton et al., 2022; Fariani et al., 2023; Sanusi et al., 2023; Zhai & Wibowo, 2023), which attracted the most research attention among studies related to AI's impact on teaching. Our synthesis revealed that many researchers (e.g., Ng et al., 2023; Pishari et al., 2020; Rizvi et al., 2023) have explored AI integration into pedagogical approaches, such as active learning, personalized learning, gaming, and collaborative project-based learning. Instead of viewing learners as passive recipients of knowledge, those various pedagogies empowered student learning and learner agency (Ouyang & Jiao,2021) and reflected "a paradigm shift from teacher-centered to learner-centered approach" in AIED (Guan et al., 2020, p. 142). With the immense power of AI on reshaping teaching and learning, instructors are challenged in transitioning their role from traditional lecturers to facilitators of personalized learning, curators of educational content, data-driven instructors, adopters of innovative pedagogies, and mentors of AI literacy (e.g., Celik et al., 2022; Chu, Hwang, Tu, & Yang, 2022; Ji et al., 2023; Ng et al., 2023; Rizvi et al., 2023). However, we found a scarcity of reviews from both educational settings in preparing instructors to help them adjust to AI-powered teaching and recalibrate their roles in the socio-technical ecosystem. This highlights the urgent need to enhance educators' competences in keeping up with AI technologies, developing AI-empowered innovative pedagogies, leveraging AI into daily teaching practices, and fostering knowledge and skills in ethical and responsible use of AIED (e.g., Gentile et al., 2023; Wang et al., 2023). To achieve these goals, educational systems should set professional standards on teacher digital competences, incentive teacher participation in professional development, and establish a wider digital education ecosystem to support teacher education, ongoing training, and innovative initiatives and projects related to digitalization (Foster, 2023).

Aside from those two commonalities in insufficient educator training and recalibration of pedagogies and instructor roles, we identified a nuanced difference in curriculum design between PreK-12 education and higher education: The former is focused on AI literacy courses (e.g., Casal-Otero et al., 2023; Sanusi et al., 2023; Yue et al., 2022) and the latter on incorporation of AI in teaching methodologies (e.g., Blumenstein, 2020). Different from AI integration to augment teaching and learning (i.e., AI for education), AI literacy, in essence, is education for AI to nurture competencies in (1) knowing and understanding fundamental AI concepts, skills, and ethics, (2) using and applying AI in various contexts and daily life, and (3) evaluating and designing/co-creating with AI applications (Ng et al., 2021). Thus, it is crucial to distinguish those two terms since they have been often mingled under the big umbrella of AIED, which might prevent scholars and educational professionals from developing a clear theoretical underpinning for AIED research.

Interestingly, our synthesis unveiled that many AI literacy curriculum or teaching units were short termed without being informed by an educational theoretical framework and implemented in informal educational settings (e.g., in labs, museums and at home). This finding confirmed that AI literacy has not been widely introduced to PreK-12 classrooms and accentuated the crucial role in adopting learning principles or theories to guide curriculum design and involving PreK-12 teachers in the process of curriculum development (Chen et al., 2020; Olari et al., 2022).

AI's impact on administration is another important thematic focus, including its potential influence on providing student performance analysis and support, enhancing administrative efficiency, and discerning research priorities. While we identified a resemblance between PreK-12 and higher education that both settings used AI to predict students' performance and provide necessary support systems (e.g., Abu Saa et al., 2019; Banihashem et al., 2022; Crescenzi-Lanna, 2020), only higher education has a heavy focus on applying AI to anticipate student dropout rates given that universities cope with the challenge of high student dropout rates (e.g., Ifenthaler & Yau, 2020; Salas-Pilco & Yang, 2022; Sghir et al., 2023). Notably, most included reviews investigating AI's impact on administration addressed how AI-solutions could facilitate student performance analysis. By contrast, less studies (e.g., Salas-Pilco & Yang, 2022; Umer et al., 2023; Zawachi-Richter et al., 2019) mainly in higher education probed how AI could increase administrative efficiency, such as automation of repetitive tasks and data analysis, personalized service, resource allocation, and cost reduction. Further, a scarcity of studies (e.g., Salas-Pilco & Yang, 2022) offered insights into harnessing AI to prioritize research areas in higher education even though research is considered as a core mission for many universities. Despite insufficient reviews probing how AI might shape research prioritization, a Nature survey asked over 1600 researchers whether they believed AI would become useful for their fields in the next ten years and found that more than half anticipated AI applications to be "very important" or "essential", indicating an upward trend for incorporating AI in empowering research (Van Noorden & Perkel, 2023).

Implications for Future Research

Based on our synthesis, we identified four major areas that demand more research attention, including conceptualization of AI and AIED, contextualization of AI use, collaboration of international multiple stakeholders, along with consolidation of methodology and quality assessment criteria.

First, a better conceptualization of AI and AIED is in dire need. Although many included reviews examined either general or specific AI applications used in education, we noticed that limited studies defined AI; and there was no consensus around a definition. The most adopted definition perceives AI as machines/technologies that emulate human intelligence or performance (McCarthy et al., 2006) without underscoring how AI, as a socio-technical ecosystem, can shape human-technology interaction and play shifting roles in education, such as AI-directed, AI-supported, and AI-empowered learning (Dignum, 2021; Ouyang & Jiao, 2021). This echoed Bilgic et al. (2022)'s concern regarding the lack of common ground and depth in defining

AI, who also called for developing functional AI definitions catered to individual subject areas. In a similar vein, AIED has been ambiguously used as an umbrella term to refer to different meanings, such as "education for AI" (i.e., education for understanding and using AI) and "AI for education" (i.e., AI integration to augment various aspects of education) (Paek & Kim, 2021). Therefore, a comprehensive conceptualization of AI and AIED should be developed, which, in turn, offers researchers a clear and consistent theoretical underpinning to study AI use in education.

Second, researchers need to strive towards a more contextualized understanding of AI and its effects on different facets of education. That is, future research should more thoroughly explore the learning contexts in which AI is applied, including but not limited to, educational levels, subject areas, student characteristics, learning environment and modalities. Compared to the numerous studies conducted in higher education, AI use in PreK-12 education is still at an early stage; and more empirical and practical research needs to probe how AI can be leveraged to enrich teaching and learning in formal education. As studies on computer science and STEM related subjects predominate research in AIED, researchers are challenged to delve into AI use in other subject areas, such as social sciences, arts, and communications. More specificity about the formal or informal environment, learning modalities, and learners' demographic characteristics is also essential for researchers to accurately decipher the impact of AI. In brief, scrutinizing contextualization will help researchers better analyze whether AI can function appropriately and effectively as intended once embedded in authentic settings (Sheikh et al., 2023).

Third, there is a need for enhancing interdisciplinary co-creation and multi-stakeholder collaboration at local, national, and international levels to more extensively study AI's promises to improve education (Bahroun et al., 2023; Bond et al., 2024; Dai et al., 2023; Molenaar & Sleegers, 2023). In particular, those collaborative efforts could be aligned with underexplored research areas as revealed by the current review. For instance, AI's influence on students' social-cultural and psychomotor developments as well as higher level cognitive and metacognitive learning skills should be better assessed given that most existing studies focused on affective and lower-level cognitive learning domains. Additionally, researchers should investigate educators' training and competencies in AI use, AI ethics, pedagogical strategies, and curriculum design which are less examined compared to the technical aspects, such as AI applications and algorithms (e.g., Bhutoria, 2022; Celik et al., 2022; Dogan et al., 2023). Future research efforts could also be geared towards evaluating how AI can enhance research prioritization due to the limited literature on this topic.

Fourth, we are calling for a consolidation of methodology and criteria for assessing the quality of systematic reviews. With exponential growth of AI studies, increasing methodology complexity, and accessible technologies for data extraction, analysis and visualization, more researchers (e.g., Han et al., 2019; Villegas-Yarlequé et al., 2023) recognized the value of employing text-mining and bibliometric analysis in conducting reviews, especially for large-scale studies. However, there is a lack of methodological guidance on how to design, implement, report, and assess systematic reviews that incorporated or completely relied on text-mining and bibliometric analysis (e.g., Lim & Kumar, 2024; Montazeri et al., 2023). As unveiled by the current study, some existing quality assessment criteria, such as inter-rater

reliability of coding and sufficient details about individual studies, do not apply to text-mining and bibliometric approaches. Further, using inter-rater reliability in an approach that does not assume objective coding, such as critical thematic analysis, would even be an indicator for low quality (Braun & Clarke, 2022). These gaps in existing research highlight a strong demand in consolidating existing systematic review methodology and corresponding criteria to evaluate the quality of reviews by taking the unique features and benefits of text-mining and bibliometric analysis into consideration.

Conclusion and Limitations

To the best of our knowledge, the current study is the first tertiary review (i.e., review of reviews) that employed a SMR approach and bibliometric analysis to examine systematic reviews on AIED, including both PreK-12 education and higher education. Our findings revealed the essential characteristics of 126 systematic reviews since 2014 as well as AI's impact on different facets of education, such as learning, teaching, and administrative and institutional service. In addition, our study shed light on commonalities and nuanced differences between higher education and PreK-12 education and how to strengthen future research on AIED.

Despite our attempt to cover as much quality research as possible, the current study has five major limitations. First, this tertiary review was focused on peer-reviewed journal articles from two high-quality databases, Scopus and WoS, and therefore did not include gray literature, conference proceedings, and relevant studies from other databases. Second, due to the SMR methodology we adopted, the current review only contained systematic reviews which excluded traditional literature review, scoping review, meta-analysis, and other types of review. Third, our study may not encompass all studies on AIED due to the lack of clear and consistent way to define and categorize AI and existence of numerous emerging AI technologies. Fourth, there might be other similarities and differences between PreK-12 education and higher education not fully captured by the current study given that many systematic reviews did not provide information regarding the educational levels and a limited number of reviews were focused on PreK-12 education. Last, this study mainly reports on the potential positive impacts of AI on education because most included reviews seldom mention the drawbacks of AI use. With more emerging systematic reviews on AIED, future research can further scrutinize how AI might negatively impact various facets of education, including teaching, learning, and administration.

Appendix

Appendix A: Features of the 126 Included Reviews

Table 5

Table 5 Note: This data is mandatory. Please provide.

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
1	Abu Saa et al. (2019)	36 research articles 2009–2018	United Arab Emirates	Data mining	Higher education	Not specified	Not specified
2	Agrusti et al. (2019)	73 documents 1999–2019	Italy	Data mining	Higher education	Not specified	Not specified
3	Ahadi et al. (2022)	991 studies Jan.2000-Jan.2022	Australia	Text mining	Not specified	Not specified	Not specified
4	Ahmad et al. (2022)	44 papers Jan. 2016 -Oct. 2021	Indonesia	Automatic content analysis, mostly machine learning	Not specified	Not specified	Multiple modalities: online & blended
5	Albreiki et al. (2021)	78 papers 2009 to 2021	United Arab Emirates	Machine learning	Multiple levels; mostly higher education	Not specified	Not specified
6	Alenezi & Faisal (2020)	30 papers 2013 to 2019	Kuwait	Machine learning	Not specified	Not specified	Online learning
7	Alfoudari et al. (2021)	105 articles between 2000 and 2019	Bahrain	Smart classrooms	Not specified	Not specified	Online learning
8	Aljarrah et al. (2021)	15 research 2007–2020	Cyprus	Textual emotion analysis	Not specified	Not specified	Online learning
9	Alonso-Fernandez et al. (2019)	87 studies 2007–2019	Spain	Game & learning analytics	Multiple levels	Various subjects; most studies on math and science	Not specified
10	Aydin et al. (2023)	26 studies 2000–2021	Turkey	Decision systems for adaptive serious games	Multiple levels	Not specified	Not specified

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
11	Aydogdu (2020)	48 studies 2007–2019	Turkey	Data mining	Multiple levels	Not specified	Not specified
12	Ba & Hu (2023)	15 studies Jan.2008-Dec. 2022	Hong Kong, China	Machine learning	Multiple levels; most studies on higher education	Not specified	Research labs, classroom environments, & out-of-classroom settings
13	Banihashem et al. (2022)	46 studies 2011 to 2022	Netherlands	Data mining	Higher education	Not specified	Online learning
14	Bhutoria (2022)	30 papers 2019–2021	India	AI-enabled personalization	Not specified	Not specified	Not specified
15	bin Mohamed et al. (2022)	20 studies 2017–2021	Malaysia	Robotics, teachable agent & autonomous agent	Not specified	Math education	Not specified
16	bin Roslan & Chen (2022)	58 articles 2015–2021	Malaysia	Data mining	Multiple levels	Not specified	Not specified
17	Blumenstein (2020)	38 studies Jan. 2011-Dec. 2019	New Zealand	Machine learning	Higher education	Various subjects, e.g., computing (generic), computer science, computing engineer, software engineering, business	Not specified
18	Bozkurt et al. (2021)	276 studies 1976–2019	Turkey	General AIED	Multiple levels	Various subjects, e.g., computer science, social sciences, engineering, & math	Not specified

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
19	Casal-Otero et al. (2023)	179 papers Oct. 2021 -Mar. 2023	Spain	AI literacy	PreK-12 education	Not specified	Not specified
20	Celik et al. (2022)	44 studies 2004–2020	Finland	General AIED	Multiple levels	Various subjects: most studies on science, technology, engineering, & math	Not specified
21	Chen et al. (2022)	22 studies 2012–2022	China	Machine learning	Not specified	Not specified	Online learning
22	Chiu (2021)	45 studies 2010–2021	Hong Kong	Machine learning	Multiple levels	Chemical education	Not specified
23	Chu, Hwang, & Tu (2022)	13 articles 2008–2021	Taiwan	AI-based robots	Multiple levels	Various subjects, e.g., computer science, engineering, language, science; most studies on language and science	Blended learning
24	Chu, Hwang, Tu, & Yang (2022)	50 studies 1996–2020	Taiwan	AI-based profiling and prediction, assessment, intelligent tutoring systems etc.	Higher education	Various subjects; most studies on engineering (including computer) courses	Not specified
25	Costa et al. (2020)	31 studies	Brazil	Machine learning	Not specified	Not specified	Online learning
26	Crescenzi-Lanna (2020)	29 documents 2015–2019	Spain	Machine learning	PreK-12 education	Not specified	Not specified

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
27	Crescenzi-Lanna (2023)	39 papers since 2018	Spain	Deep learning, machine learning, etc.	PreK-12 education	Not specified	Not specified
28	Crompton et al. (2022)	169 studies 2011–2021	USA	AI-based gaming, intelligence tutoring etc.	PreK-12 education	Various subjects; most studies on science & math	Not specified
29	Cuellar-Rojas et al. (2022)	163 papers 1979–2021	Mexico	Intelligent tutoring systems	Not specified	Not specified	Not specified
30	da Silva et al. (2023)	16 studies 2015 to 2020	Brazil	Educational recommender systems; machine learning	Not specified	Not specified	Not specified
31	Dai & Ke (2022)	59 studies 1998–2021	USA	AI applications in simulation-based learning; machine learning	Not specified	Not specified	Online learning
32	Daoudi (2022)	80 papers 2011–2021	Tunisia	Game learning analytics; machine learning	Not specified	Not specified	Formal learning
33	Deeva et al. (2021)	109 papers 2008–2019	Belgium	Automated feedback systems	Multiple levels	Various subjects, e.g., STEM, languages, medicines, art, & special education	Multiple Modalities: online & blended; Formal learning

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
34	Dogan et al. (2023)	276 papers 1999–2022	Turkey	AI-based personalized learning, data mining and learning analytics	Not specified	Various subjects: most studies on computer science and engineering, followed by social sciences	Online learning
35	Durso & Arruda (2022)	63 studies 1994–2020	Brazil	Intelligent tutoring systems, adaptive systems, assessment, profiling and prediction etc.	Higher education	Various subjects; most studies on computer science and engineering	Online learning
36	Fang et al. (2023)	27 studies 2018-Oct. 2022	Hong Kong, China	AI story-writing	Multiple levels	Not specified	Not specified
37	Fariani et al. (2023)	39 articles 2017–2021	Indonesia	Personalized learning; machine learning	Higher education	Not specified	Not specified
38	Feng & Law (2021)	1830 articles 2010–2019	Hong Kong, China	Intelligent tutoring systems, massive open online courses, data mining, learning analytics, natural language processing, machine learning etc.	Not specified	Not specified	Not specified

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
39	Gentile et al. (2023)	102 articles 2014–2023	Italy	AI, deep learning, machine learning, natural language processing	Not specified	Not specified	Not specified
40	Gonzalez-Calatayud et al. (2021)	22 studies 2010–2020	Spain	AI-based assessments	Multiple levels	Various subjects, including computer science, math, physics, business, education, English language, medicine, statistics etc.	Not specified
41	Grimalt-Alvaro & Usart (2023)	22 studies 2012–2021	Spain	Sentiment analysis	Higher education	Various subjects: most studies on computer sciences, followed by science, engineering, and teacher training	Multiple modalities: online learning & blended
42	Hopcan et al. (2022)	29 studies 2008–2020	Turkey	AI software	Multiple levels	Various subjects, mostly in skill development, subject neutrality, and literacy, with fewer in STEM and language education	Formal & informal learning
43	Huang et al. (2022)	25 studies 2008–2021	Hong Kong, China	Chatbots	Multiple levels; most studies in higher education	Language literature	Not specified

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
44	Hwang & Chang (2021)	19 articles 2005–2020	Taiwan	Chatbots	Not specified	Various subjects: most studies on languages, followed by engineering or computers	Not specified
45	Hwang & Tu (2021)	43 articles 1996–2020	Taiwan	Intelligent tutoring systems, profiling and prediction and adaptive systems; machine learning	Multiple levels	Math education	Not specified
46	Hwang et al. (2022)	112 research articles 1993–2020	Taiwan	Intelligent agent; profiling and prediction	Not specified	Nursing education	Not specified
47	Ifenthaler & Yau (2020)	46 articles January 2013 and December 2018	Germany	Learning analytics	Higher education	Various subjects, including engineering, STEM, informatics, math, computer science etc.	Online learning
48	Ismail et al. (2023)	72 studies 2000–2020	United Arab Emirates	Personalized learning software systems	Not specified	Not specified	Online learning; formal & informal learning
49	Jeon et al. (2023)	32 studies 2010–2022	USA	Speech-recognition chatbots	Multiple levels; most studies in higher education	Language education	Formal & informal learning

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
50	Ji et al. (2023)	24 studies 2015–2021	USA	Conversational AI	Multiple levels; most studies in higher education	Language education	Formal & informal learning
51	Kaddoura et al. (2022)	135 studies before 2023	United Arab Emirates	Machine learning	Multiple levels	Not specified	Online learning
52	Kang (2023)	82 studies 2010–2022	USA	Personalizing learning, intelligent learning systems etc.	Higher education (Adult education)	Not specified	Not specified
53	Karimah & Hasegawa (2022)	47 articles 2010–2022	Japan	Automatic engagement recognition; machine learning; deep learning	Not specified	Not specified	Online learning
54	Khakpour & Colomo-Palacios (2021)	32 studies 2014–2019	Norway	Machine learning	Not specified	Not specified	Online learning
55	Khaldi et al. (2023)	90 articles 2010–2021	Algeria	Gamification; machine learning	Higher Education	Not specified	Online learning
56	Khan & Ghosh (2021)	140 studies 2000–2018	India	Data mining	Not specified	Not specified	Formal learning
57	Khanal et al. (2020)	35 papers 2016–2018	Australia	Machine learning based recommendation systems	Not specified	Not specified	Online learning
58	Klimova et al. (2023)	13 articles Jan.2016-May 2022	Czech Republic	Neural machine translation	Multiple levels; most studies in higher education	Foreign language education	Not specified

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
59	Klimova et al. (2023)	14 studies January 2018 to December 2022	Czech Republic	AI-based chatbot, virtual reality, machine learning etc.	Higher education	Teaching English as an applied language	Not specified
60	Kuhail et al. (2023)	36 studies 2011–2021	United Arab Emirates	Chatbots	Not specified	Various subjects: most studies used chatbots to teach computer science	Online learning
61	Lee (2023)	87 studies 2000–2019	Korea	Machine translation	Multiple levels; most studies in higher education	Foreign language education	Not specified
62	Ley et al. (2023)	42 studies 2004–2021	Austria	Intelligent learning technology & analytics	Multiple levels; most studies in higher education	Various subjects, including math, informatics, and foreign language learning	Not specified
63	Li et al. (2021)	7880 papers 1971–2020	China	Adaptive learning systems	Not specified	Not specified	Not specified
64	Liang et al. (2021)	71 articles 1990 to 2020	Taiwan	Intelligent tutoring system, natural language processing, data mining, machine learning, natural language parsing etc.	Multiple levels; most studies in higher education	Language education	Not specified

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
65	Lin et al. (2023)	29 papers 2014–2023	Taiwan	Intelligent tutoring systems	Multiple levels, most studies in higher education	Not specified	Formal learning
66	Martínez-Comesaña et al. (2023)	9 studies 2010–2023	Spain	AI-based Assessment	PreK-12 education	Not specified	Not specified
67	Martinez-Roig et al. (2023)	4 articles 2019–2022	Spain	Social robotics	PreK-12 education	Music education	Formal learning
68	Moore et al. (2023)	12 articles 2014–2021	USA	Machine intelligence	PreK-12 education	Various subjects: most studies on math, followed by science and language arts	Formal & informal learning
69	Moreno-Marcos et al. (2018)	88 articles 2014–2017	Spain	Machine learning	Not specified	Various subjects, including humanities, social sciences, natural sciences, formal sciences, and professions and applied sciences	Online learning
70	Mousavinasab et al. (2021)	53 papers 2007–2017	Iran	Intelligent tutoring systems	Multiple levels	Various subjects; 37.7% of the studies on computer sciences	Online learning

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
71	Ng et al. (2023)	49 studies 2000–2020	Hong Kong, China	AI-based platforms for software development, robotics, and intelligent agents	Multiple levels; most studies in higher education	Various subjects, including computer science and non-computer science majors (e.g., history, music, and education)	Not specified
72	Nigam et al. (2021)	43 studies 2015–2021	India	AI-based proctoring systems	Not specified	Not specified	Not specified
73	Normadhi et al. (2019)	78 studies 2010–2017	Malaysia	Adaptive learning; machine learning	Not specified	Not specified	Online learning
74	Okewu et al. (2021)	190 studies 2010–2018	Nigeria	Artificial neural networks; data mining	Higher education	Not specified	Not specified
75	Okonkwo & Ade-Ibijola (2021)	53 articles 1966–2019	South Africa	Chatbots	Not specified	Not specified	Not specified
76	Ongoro & Fangjiang (2023)	50 studies Jan.2010-Apr.2022	Taiwan	Digital game-based technology	PreK-12 education	English as a foreign language	Not specified
77	Ouyang et al. (2022)	32 articles 2011–2020	China	AI-based prediction, resource recommendation, assessment etc.	Higher education	Not specified	Online learning
78	Papadopoulos et al. (2020)	21 studies 2004–2018	UK	Socially assistive robots	PreK-12 education	Various subjects, e.g., math, science, & language learning	Formal learning

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
79	Papamitsiou & Economides (2014)	40 studies 2008–2013	Greece	Learning analytics; data mining	Not specified	Not specified	Online learning
80	Pishtari et al. (2020)	54 studies 2008–2019	Estonia	Learning analytics	Multiple levels; most studies in higher education	Language learning	Online learning
81	Pradana et al. (2023)	74 studies 2022–2023	Indonesia	ChatGPT	Not specified	Not specified	Not specified
82	Prinsloo et al. (2023)	192 articles 2011-February 2023	South Africa	Multimodal learning analytics	Multiple levels; most studies in higher education	Not specified	Not specified
83	Rahayu et al. (2023)	9 studies 2010–2020	Indonesia	Learning path recommender systems	Not specified	Various subjects, e.g., studying cultural heritage or law, business, & computer	Online learning
84	Raj & Renumol (2022)	52 studies 2015–2020	India	Adaptive content recommenders; machine learning	Not specified	Not specified	Online learning
85	Ramadhan et al. (2023)	32 articles 1994–2021	Indonesia	Intelligent tutoring system; gamification	Not specified	Various subjects, e.g., STEM and Non-STEM majors	Not specified
86	Ramaswami et al. (2023)	15 papers 2012–2021	New Zealand	machine learning; predictive analytics	Not specified	Not specified	Online learning
87	Rice & Dunn (2023)	18 studies 2009–2022	USA	AI-based assistive technology	PreK-12 education	Special education	Not specified

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
88	Rizvi et al. (2023)	28 studies 2019–2022	UK	Generative AI	PreK-12 education	Not specified	Not specified
89	Salas-Pilco, Xiao, & Hu (2022)	30 studies 2017–2021	China	Learning analytics; machine learning	Multiple levels	Teacher education	Not specified
90	Salas-Pilco, Xiao, & Oshima (2022)	27 articles 2017–2021	China	Machine learning, mobile technology, serious games, learning analytics, virtual/augmented reality and robotics	Multiple levels, most studies in higher education	Various subjects, e.g., STEM and engineering	Not specified
91	Salas-Pilco & Yang (2022)	31 articles 2016–2021	China	Machine learning, deep learning, natural language processing etc.	Higher education	Not specified	Not specified
92	Sallam (2023)	60 records 2022–2023	Jordan	ChatGPT	Not specified	Health education	Not specified
93	Santomartino & Yi (2022)	19 studies 2018–2021	USA	AI-based voice recognition software	Not specified	Radiology & medical education	Online learning
94	Santos et al. (2022)	17 studies 2018–2021	Brazil	AI and machine learning	Higher education	Health sciences	Not specified
95	Sanusi et al. (2023)	43 studies 2010–2021	Finland	Machine learning	PreK-12 education	Not specified	Formal & informal learning
96	Sekeroglu et al. (2021)	176 studies 010–2020	Cyprus	Predictive learning analytics; machine learning	Multiple levels; most studies in higher education	Not specified	Not specified

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
97	Sghir et al. (2023)	74 studies 2015–2022	Morocco	Predictive learning analytics	Higher education	Various subjects; most studies on education & computer science	Not specified
98	Sharadgah & Sa'di (2022)	64 articles 2015–2021	Saudi Arabia	Novel AI-based systems	Not specified	English language teaching and learning	Not specified
99	Shin & Shim (2021)	64 articles 2010–2019	South Korea	Data mining	Not specified	Math & science education	Not specified
100	Sikstrom et al. (2022)	31 studies 2010–2020	Finland	Pedagogical agents	Not specified	Not specified	Not specified
101	Silva et al. (2018)	80 studies 2002–2017	Brazil	Automatic question classifiers; machine learning	Not specified	Not specified	Not specified
102	Souabi et al. (2021)	51 articles 2007-first part of 2021	Morocco	Recommendation systems	Not specified	Not specified	Online learning
103	Sulistio et al. (2023)	48 studies 2009–2022	Indonesia	Machine learning	Not specified	Studying Hadith	Not specified
104	Sun et al. (2023)	19 studies 2016–2021	Finland	Intelligent game-based learning	Not specified	Not specified	Online learning
105	Tabuenca et al. (2021)	68 papers 2000–2019	Spain	Machine learning, learning analytics, data mining etc.	Not specified	Not specified	Online learning
106	Tahiru (2021)	23 articles 2010–2019	Ghana	AI/ intelligent tutoring system	Not specified	Not specified	Not specified

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
107	Tan et al. (2022)	41 articles Jan. 2002- May 2022	Singapore	AI techniques for collaborative learning	Multiple levels; most studies in higher education	Not specified	Multiple modalities: online & f2f
108	Tang et al. (2023)	86 papers 1998–2019	Taiwan	AI-supported e-learning; intelligent tutoring systems	Not specified	Various subjects; most studies in computer engineering, computer science & nature science	Online learning
109	Topali et al. (2023)	38 articles 2014–2022	Spain	Learning analytics	Not specified	Various subjects; most studies on formal sciences (i.e., programming & math)	Online learning
110	Umer et al. (2023)	40 studies 2008–2018	Australia	Predictive analytics	Higher education	Not specified	Not specified
111	Wang et al. (2023)	40 studies 2011–2022	China	Intelligent tutoring systems	Multiple levels	Various subjects, e.g., language, math, science, computer science, medicine, history, economics, geometry, and engineering	Not specified
112	Weng & Chiu (2023)	83 studies	Hong Kong	Intelligent computer assisted language learning	Not specified	Language learning	Online learning

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
113	Xu & Ouyang (2022a)	51 studies Jan.,2005 -Aug., 2021	China	AI -based instructional and learning process	Not specified	Not specified	Not specified
114	Xu & Ouyang (2022b)	63 articles 2011–2021	China	AI-based learning prediction, intelligent tutoring systems, automation, robots	Multiple levels	Various STEM subject, e.g., science, technology, engineering, math, & cross-disciplinary	Multiple modalities: online & f2f; informal learning
115	Yan et al. (2023)	118 papers since 2017	Australia	Large language model-based innovations	Not specified	Not specified	Not specified
116	Yang & Kyun (2022)	25 papers January 2007-February 2021	Australia	AI-supported automatic writing evaluation systems, robots, agents, intelligent tutorial systems, & voice-based smartphone apps.	Multiple levels; most studies in higher education	Language learning	Not specified
117	Younis et al. (2023)	82 articles 2014–2023	Malaysia	Educational robots & natural language processing	Multiple levels	Various subjects, e.g., programing and arithmetic	Not specified
118	Yue et al. (2022)	32 studies 2010–2022	Hong Kong, China	machine learning; AI literacy	PreK-12 education	AI teaching/education	Multiple modalities: online, blended & others; formal & informal learning

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
119	Zafari et al. (2022)	210 studies 2011–2021	Iran	Machine learning; intelligent tutoring systems	PreK-12 education	Various subjects, e.g., math, geometry, algebra, arithmetic, science, physics, chemistry, geography, biology, microbiology, robotics, programming, engineering, language learning, history, music	Multiple modalities: online & animation; formal & informal learning
120	Zawacki-Richter et al. (2019)	146 articles 2007–2018	Germany	Intelligent tutoring systems, adaptive systems, assessment and evaluation and profiling and prediction	Higher education	Various subjects; most studies on computer science and STEM	Not specified
121	Zhai & Wibowo (2022)	48 articles 2012–2022	Australia	Chatbots	Not specified	Second language acquisition	Not specified
122	Zhai & Wibowo (2023)	28 articles 2013–2022	Australia	AI dialogue systems	Higher education	English as foreign language	Not specified
123	Zhai et al. (2020)	49 articles	USA	Machine learning	Multiple levels; most studies in higher education	Various science subjects	Not specified

Table 5 (continued)

NO.	Author/Year	Review Size & Publication Years	Country of 1st Author	AI Applications/ Algorithms	Educational Level(s)	Subject Area(s)	Learning Environments/ Modalities
124	Zhai et al. (2021)	100 papers 2010–2020	China	Intelligent tutoring systems, assessment systems, intelligent virtual lab, chatbot, robotics etc.	Multiple levels; most studies in higher education	Various subjects; most studies on science, humanities & social sciences	Online learning
125	Zhan et al. (2022)	125 papers 1999–2021	China	Game-based learning	Multiple levels; most studies in higher education	AI teaching/education	Online learning
126	Zhu et al. (2022)	166 articles 2011–2021	USA	Learning analytics	Not specified	Various subjects; most studies on computer science	Not specified

Table 6 Note: This data is mandatory. Please provide.

Themes	Subthemes	Examples
Impact on student learning	Effects on domains of learning	"[Pedagogical agent]s' communication has a positive effect on learners' motivation, academic success, and cognitive load" (Sikstrom et al., 2022, p. 9). "The reviewed papers reported an improvement in affective outcomes (better perception, improved attitude), as the participation left students more enthusiastic and motivated to explore further AI topics" (Rizvi et al., 2023, p. 7). "The study reached positive results, showing that the intervention enhances mathematical concepts and arithmetic thinking and improves students' thinking skills" (Younis et al., 2023, p. 7).
	AI-powered adaptive & personalized learning	"Some benefits of AI in the learning process include: helping students learn more effectively by providing personalized feedback and recommendations" (Moore et al., 2023, p. 15). "it is considered that AI and ML algorithms could help to personalize learning according to the students' needs" (Salas-Pilco, Xiao, & Oshima, 2022, p. 7). "AI, coupled with the existing technological involvement in the sector, is directed to systematically modify and redesign certain crucial components of the education system to capture and accommodate individual-specific factors" (Bhutoria, 2022, p. 5).
	AI-based assessments of student learning	"NLP [Natural Language Processing] is used to identify the correct and incorrect statements in an assignment by combining extraction rules and machine learning based information extractors" (Alenezi & Faisal, 2020, p. 2979). "The main contributions of the application of AI in the assessment of students at these lower educational levels focus on predicting their performance, automating and making evaluations more objective by means of neural networks or natural language processing, the use of educational robots to analyse their learning process, and the detection of specific factors that make classes more attractive" (Martínez-Comesaña et al., 2023, p. 90). "The review showed that the field of automatic writing evaluation systems has been a significant area of research and AI-supported automatic writing evaluation is an emerging part of this area" (Yang & Kyun, 2022, p. 189).

Table 6 (continued)

Themes	Subthemes	Examples
Impact on teaching	Instructional pedagogies & instructor role recalibration	"Regarding the methods and strategies implemented by teachers in educational processes, the main contribution of AI relays the shift in the center of gravity of teaching processes and models: from teacher-centered models and processes to learner-centered models and processes" (Gentile et al., 2023, p. 7). "For instance, some teaching agents recommended tutorials to students based upon learning styles" (Kuhail et al., 2023, p. 990). "These studies involved the use of AI in activities focused on using a specific pedagogical approach, such as active learning, personalized learning, gaming, and collaboration. These included activities led by the educator in person as well as the virtual educators (e.g., the intelligent tutors)" (Crompton et al., 2022, p. 9).
	Curriculum design	"The LALGD model is aimed at encouraging dialogue among key stakeholders involved in the planning and development of curriculum, to help design learning for student success" (Blumenstein, 2020, p. 25). "The duration of the teaching units varies considerably. The teaching units can involve short and focused activities (minimum lesson duration: <1 h), multiple extracurricular lessons (3–15 h in total), and long-term courses (maximum lesson duration: ~80 h). However, most are relatively short-term courses (Fig. 8). Half of the teaching units (16 of 32) in the reviewed studies comprise only one session lasting no more than 3 h, suggesting that a short lesson may be sufficient for children to develop an understanding of AI concepts" (Yue et al, 2022, p. 11). "Approaches to curriculum development differ widely, ranging from the product-centered model (technical-scientific perspective) to the process-centered model (learner perspective) (Yue et al., 2022). AI literacy can be launched in secondary education depending on the age and computer literacy of the students" (Casal-Otero et al., 2023, p. 10).

Table 6 (continued)

Themes	Subthemes	Examples
	Teacher professional development	"Four articles were identified to have specifically focused on teacher training and professional development on AI and ML" (Sanusi et al., 2023, p. 5984). "Many teachers, especially those in primary and secondary settings, reported a lack of confidence in their own understanding because the field of AI is changing rapidly, and they often lack access to relevant technological pedagogical content knowledge" (Ng et al., 2023).
Impact on administration	Student performance analysis & support	"data mining techniques are used for predicting university dropout. We identified 6 data mining classification techniques, 53 data mining algorithms and 14 data mining tools" (Agrusti et al., 2019, p. 161). "Most studies utilized logs of these interactions as absolute counts for making onward predictions, without normalizing them by representing values for each student in relation to those of their cohort" (Ramaswami et al., 2023, 971). "predictive models can improve the process of enrolling by analyzing students' data to assist them in their orientation thus improving their chances of acceptance. Studies addressing enrollment focus either on the admission test, the submission application, or both" (Sghir et al., 2023, p. 8312).
	Administrative efficiency	"The Chatbot will function as a digital assistant, advising tertiary institutions on how to improve their current services, generate new innovative ideas, and cut labor costs" (Okonkwo & Ade-Ibijola, 2021, p. 6). "Predictive modeling is beneficial because it gives accurate insight into educational questions and allows decisions to be made in advance; it is also time-saving and cost-effective" (Salas-Pilco & Yang, 2022, p. 7). "the research shows that admission decisions can be predicted at high levels of accuracy, so that an AI solution could relieves the administrative staff and allows them to focus on the more difficult cases" (Zawacki-Richter et al., 2019, p. 12).
	Research priorities	"The use of AI to detect scientific communities and prominent research areas can provide important information for policymakers as they look to set research priorities" (Salas-Pilco & Yang, 2022, p. 7).

Appendix B: Code Book with Illustrative Examples

Table 6

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